



MICHIGAN CENTER FOR REGENERATIVE MEDICINE

WHY COMPOUNDING MATTERS

QUALITY. IMPROVING LIVES.

WHAT IS A COMPOUNDING PHARMACY?

Compounding pharmacies blend the art and science of preparing customized medications tailored to meet specific patient needs. Licensed pharmacists work in regulated environments to create formulations that aren't available commercially, ensuring medications are as effective and well-tolerated as possible.

Think of a compounding pharmacy as a bridge between traditional pharmacy care and personalized healthcare, addressing gaps in standard treatments to support proactive, preventative medicine.



A HISTORY OF PATIENT CARE

MIDDLE AGES TO 19TH CENTURY:

Apothecaries, the original pharmacists, hand-crafted all medicines using mortars, pestles, and herbal expertise. Each remedy was customized for the patient's unique symptoms and needs.

MID-20TH CENTURY SHIFT:

Mass manufacturing of pharmaceuticals led to standardized drugs, reducing the need for compounding as pre-made options became efficient and widely available.

EARLY 1900S:

Nearly every prescription was compounded. Pharmacists were central to care, mixing raw ingredients into personalized doses.

MODERN REVIVAL:

In recent times, compounding has seen rising demand for personalized medicine in integrative and functional approaches: Focusing on unique treatment plans, preventative care, and proactive health strategies.

COMPOUNDING PHARMACIES TODAY: WHO BENEFITS?

Modern compounding pharmacies operate under strict regulations to deliver safe, high-quality medications. They cater to diverse needs, including:

- Patients allergic to dyes, fillers, or preservatives in commercial drugs.
- Requirements for alternative administration forms
- Custom concentrations or combinations for specialized therapies.
- Preventative and proactive healthcare tailored to individual lifestyles and goals.



By partnering with compounding pharmacies, providers/clinics can enhance patient adherence, seek alternative solutions, and foster long-term health plans, ultimately strengthening your practice's reputation for innovative care.

UNDERSTANDING 503-A VS. 503-B

Compounding pharmacies fall into two main categories, each designed for different scales of need.

503-A Compounding Pharmacy: A patient-specific compounding pharmacy that compounds medications only after receiving a prescription for an individual patient, focusing on custom formulations tailored to unique patient need

503-B Compounding Pharmacy: An outsourcing facility that mass-produce compounded medications for hospitals, clinics, and providers without requiring individual patient prescriptions.

A SIMPLE APPROACH

Imagine **503-A compounding pharmacies** as a boutique bakery crafting a cake exactly to your order with specific flavors, size, and decorations tailored just for you. In contrast, **503-B compounding pharmacies** are like a commercial bakery producing thousands of standardized cakes for stores: reliable and efficient, but not customized.



<u>Aspect</u>	503-A Compounding Pharmacy	503-B Compounding Pharmacy
<u>Focus</u>	Patient-specific prescriptions only	Mass production without individual prescriptions
<u>Regulations</u>	Regulated by State Boards of Pharmacy and follows USP <795> and <797> standards	Follows Current Good Manufacturing Practices (cGMP)
<u>Typical Users</u>	Individual patients via clinics for custom care	Hospitals, clinics, and healthcare systems for bulk needs
<u>Key Strength</u>	Highly personalized formulations	Consistent, large-scale supply with rigorous quality controls



COMPOUNDING PHARMACIES PRIORITIZING SAFETY AND PRECISION

While various sources exist for wellness products, understanding the differences helps providers make informed choices that align with regulatory standards and patient safety.

COMPOUNDING PHARMACIES:



- ✓ Licensed and regulated by state pharmacy boards
- ✓ Use only USP-grade active pharmaceutical ingredients (APIs) and excipients, verified for identity, purity, and potency.
- ✓ Adhere to USP <795>, <797>, and <800> guidelines for sterility, stability, and safe handling procedures.
- ✓ Provide patient-specific or clinic-use medications with full traceability, documentation, and quality assurance.
- ✓ Enable clinical partnerships, allowing providers to maintain oversight and customize treatments.

RESEARCH-GRADE OR NUTRACEUTICAL PRODUCTS:



- ✗ Often produced in labs not licensed as pharmacies, with lighter (or no) regulation focused on research or supplement standards rather than human drug compounding.
- ✗ May use ingredients that don't meet pharmaceutical-grade purity, labeled for "research use only" or "not for human consumption" to navigate regulations.
- ✗ Typically lack the same level of sterility controls, testing, and traceability, which can introduce variability in bioavailability or dosing.
- ✗ Not designed for dispensing as medications, limiting their role in clinical settings without additional verification.

WHAT THIS MEANS FOR YOU

By choosing compounding pharmacies, clinics gain peace of mind through regulated processes that minimize risks and maximize efficacy. Alternatives can play a role in exploratory or non-human contexts, but compounding ensures medications are crafted with clinical precision: helping you avoid potential inconsistencies and support evidence-based care when optimizing patient therapies.

While research-grade products spark innovation, compounding pharmacies translate that into safe, patient-ready therapies under regulatory oversight. Which means partnering with compounding pharmacies, like VPI Compounding, gives you control over formulations and product offerings, reducing reliance on variable sources and ensure the highest quality standard for your patient care.

KEY TAKEAWAY: ELEVATING PATIENT CARE THROUGH TRUSTED PARTNERSHIPS

Compounding pharmacies uphold the highest standards of precision, quality, and safety, making them ideal for personalized medicine. They empower providers to address unique patient needs without compromising on oversight or efficacy. In a world where patients increasingly seek tailored solutions, relying on regulated compounding helps prevent gaps in care, such as patients turning to unregulated online sources.

The result? Stronger patient outcomes, reduced liabilities, greater trust, and a proactive approach to health that sets your clinic apart.





MICHIGAN CENTER FOR
REGENERATIVE MEDICINE



CELLULAR HEALTH

Cellular health drugs are medications designed to improve and maintain the function of cells, which are the building blocks of all tissues and organs in the body. These drugs target various aspects of cellular health, such as energy production, oxidative stress reduction, and cell regeneration. By optimizing cellular function, they help protect against diseases, slow aging processes, and enhance overall physical and mental well-being.

One of the primary benefits of cellular health drugs is their role in enhancing mitochondrial function, the part of the cell responsible for energy production. Cellular health drugs also focus on reducing oxidative stress, which can damage cells and contribute to aging and diseases like cancer, heart disease, and neurodegenerative disorders.

In summary, cellular health drugs support the body's basic functions at the cellular level, improving energy, reducing the risk of disease, and promoting faster recovery. These benefits contribute to better overall health, longevity, and quality of life.

CELLULAR HEALTH



Lyophilized Powder
Nasal Spray
SQ Injectable
Topical Cream

NAD+

NAD⁺, which stands for Nicotinamide Adenine Dinucleotide, is a critical coenzyme found in every cell of the body, involved in various metabolic processes, including energy production, DNA repair, and cell signaling. As we age, NAD⁺ levels decline, leading to a decrease in cellular function and an increase in age-related diseases. Supplementing with NAD⁺ has been shown to enhance mitochondrial function, improve energy metabolism, and promote longevity. Clinically, NAD⁺ is used in anti-aging therapies, addiction recovery, and neurodegenerative disease treatment. Its role in maintaining cellular health and promoting longevity makes NAD⁺ a cornerstone in the field of regenerative medicine.

Suggested Dosing Information

IM or SQ: [100mg - 200mg] 3 - 5 days per week
Nasal: 75mg - 150mg per day
Topical: 25mg-100mg per day

Benefits of NAD+

1. Boosts Energy Metabolism: NAD⁺ is pivotal in the mitochondria, the energy powerhouses of cells, helping to convert food into the energy that powers cells.
2. Promotes Cellular Repair: NAD⁺ is involved in DNA repair and the maintenance of cellular health. It supports the activity of sirtuins, a family of proteins that regulate cellular health and aging, and PARPs (poly ADP-ribose polymerases), which repair damaged DNA.
3. Regulates Circadian Rhythms: NAD⁺ levels fluctuate throughout the day and influence the circadian clocks that govern daily physiological cycles. Maintaining robust NAD⁺ levels helps sustain a healthy sleep-wake cycle.
4. Supports Brain Health: NAD⁺ has been shown to benefit brain health by supporting the function of neurons and protecting against neurodegenerative diseases. It helps maintain mitochondrial function in brain cells, crucial for preventing cognitive decline.
5. Anti-Aging Effects: Enhancing NAD⁺ levels can stimulate the activity of sirtuins and other anti-aging molecules, potentially slowing down the aging process and extending lifespan.
6. Enhances Weight Management: By improving the efficiency of your metabolism, higher levels of NAD⁺ can help manage body weight and prevent metabolic disorders such as diabetes.

CELLULAR HEALTH



Capsule
Nasal Spray

Nicotinamide Mononucleotide (NMN)

Nicotinamide Mononucleotide (NMN) is a precursor to NAD⁺ and plays a crucial role in the body's energy production and metabolic processes. NMN is converted into NAD⁺ within cells, supporting mitochondrial function and enhancing overall cellular energy. As NAD⁺ levels naturally decline with age, NMN supplementation is believed to help restore youthful energy levels, improve metabolic health, and promote longevity. Clinically, NMN is being studied for its potential to combat age-related decline, improve insulin sensitivity, and enhance cognitive function. Its role in NAD⁺ biosynthesis makes NMN a promising candidate in the field of anti-aging and metabolic health.

Suggested Dosing Information

300mg PO daily

Benefits of NMN

1. Supports Cellular Energy Production: NMN boosts NAD⁺ levels, which are vital for energy production in cells, helping to improve overall energy levels and vitality.
2. Enhances Mitochondrial Function: By increasing NAD⁺, NMN supports mitochondrial health, the powerhouses of cells, leading to more efficient energy utilization and improved cellular function.
3. Reduces Age-Related Decline: NMN helps combat the natural decline in NAD⁺ levels that occurs with aging, supporting healthy aging and reducing the effects of age-related diseases.
4. Supports Overall Metabolic Health: NMN plays a role in maintaining healthy metabolism, aiding in processes such as fat breakdown, insulin sensitivity, and glucose regulation, which are important for metabolic health.
5. Improves Cognitive Function: By supporting mitochondrial function and energy production in the brain, NMN may improve cognitive performance, memory, and focus, especially as we age.
6. Aids in Extending Lifespan: NMN's ability to increase NAD⁺ levels and promote cellular repair and regeneration may contribute to longevity and extended lifespan in some animal studies, with potential for humans.
7. Supports Overall Vitality & Mood: NMN helps enhance mood and vitality by improving energy levels and metabolic function, contributing to a greater sense of well-being.
8. Enhances Cellular Repair: NMN supports the repair of damaged DNA and improves cellular regeneration, which is critical for maintaining healthy tissues and slowing the aging process.



COGNITIVE MODULATOR

Cognitive modulator drugs are medications designed to enhance or regulate cognitive functions such as memory, attention, focus, and mental clarity. These drugs target various neurotransmitters in the brain, such as dopamine, serotonin, and acetylcholine, which are responsible for cognitive processes.

One of the key benefits of cognitive modulator drugs is their ability to improve focus and attention, particularly in individuals with ADHD, enabling them to perform better in school, work, and daily tasks.

These drugs also help enhance memory and learning, which is beneficial for those with age-related cognitive decline or neurodegenerative conditions like Alzheimer's. By boosting cognitive performance, these medications can help individuals maintain independence and mental sharpness as they age.

In summary, cognitive modulator drugs play an important role in improving memory, focus, and mental performance, particularly for those with cognitive impairments, leading to enhanced productivity, independence, and well-being.

COGNITIVE MODULATOR

ALPHA-GPC

Nasal Spray

Alpha-GPC

Alpha-GPC (L-alpha-glycerylphosphorylcholine) is a cholinergic compound derived from phosphatidylcholine, known for its cognitive-enhancing properties. It functions as a precursor to acetylcholine, a neurotransmitter crucial for memory, learning, and overall cognitive function. Clinically, Alpha-GPC is used to improve cognitive performance, particularly in conditions like Alzheimer's disease and dementia. It is also popular in sports medicine for its potential to enhance power output and recovery. Its ability to cross the blood-brain barrier effectively makes Alpha-GPC a valuable supplement for both cognitive health and physical performance.

Suggested Dosing Information

18.75mg - 37.5mg daily

Benefits of Alpha-GPC

1. Enhanced Cognitive Function: Alpha-GPC is reputed to improve memory, thinking skills, and learning. It is particularly popular among students and professionals looking to enhance mental performance.
2. Increased Acetylcholine Levels: By providing choline to the brain, Alpha-GPC supports the production of acetylcholine, essential for various brain functions, including memory formation and muscle control.
3. Neuroprotective Properties: Some studies suggest that Alpha-GPC can help maintain brain health, potentially providing benefits in neurodegenerative conditions such as Alzheimer's disease and dementia. It may help mitigate the cognitive decline associated with these conditions.
4. Supports Athletic Performance: Alpha-GPC has been shown to increase power output in athletes, possibly due to its role in facilitating neurotransmission and enhancing growth hormone production.
5. Improvement in Stroke Recovery: There is evidence that Alpha-GPC can aid in the recovery of brain function after an ischemic stroke when used as part of early treatment.
6. Mood Regulation: The increase in cholinergic activity might also help with mood stabilization, potentially alleviating symptoms of mood disorders.

COGNITIVE MODULATOR



Nasal Spray

PE 22-28

PE-22-28, also known as Mini-Spadin, is a neuropeptide derived from the larger peptide VIP (vasoactive intestinal peptide), known for its neuroprotective and anti-inflammatory properties. It works by modulating the activity of the immune system and protecting neurons from damage. PE 22-28 is being studied for its potential in treating neurodegenerative diseases such as Alzheimer's disease and for its ability to enhance cognitive function. Its ability to protect neurons and reduce inflammation makes PE 22-28 a promising candidate in the field of neuroprotection and cognitive enhancement.

Suggested Dosing Information

100mcg - 200mcg daily in the morning

Benefits of PE 22-28

1. Supports Mood Regulation: PE-22-28 can help stabilize mood by influencing neurotransmitter pathways, leading to more balanced emotional states.
2. Reduces Symptoms of Depression: It has shown potential in alleviating depressive symptoms by promoting positive mood and reducing negative emotional states.
3. Enhances Cognitive Function: PE-22-28 may support cognitive processes such as memory, learning, and mental clarity, contributing to improved brain function.
4. Improves Overall Mental Well-Being: By enhancing mood and cognitive abilities, PE-22-28 promotes an overall sense of mental well-being and emotional balance.
5. Aids in Anxiety Disorders: PE-22-28 can help reduce anxiety and stress, providing relief for individuals dealing with anxiety disorders and enhancing emotional calmness.
6. Supports Emotional Resilience: By improving mood regulation and reducing symptoms of anxiety and depression, PE-22-28 fosters emotional resilience, helping individuals better cope with stress and adversity.
7. Improves Focus & Clarity: This peptide can enhance mental focus and clarity, contributing to better performance in tasks that require concentration and cognitive sharpness.
8. Enhances Overall Psychological Health: Through its influence on neurotransmitter systems, PE-22-28 contributes to enhanced mental health, emotional stability, and cognitive well-being.

COGNITIVE MODULATOR



**Lyophilized Powder
Nasal Spray
SQ Injectable**

Pinealon

Pinealon is a tripeptide derived from the pineal gland, known for its role in regulating the body's circadian rhythms and promoting mental clarity. It works by modulating the activity of neurotransmitters and protecting neurons from oxidative stress. Pinealon is primarily used to improve cognitive function, enhance memory, and protect against neurodegenerative diseases. Its ability to support brain health and regulate sleep patterns makes Pinealon a promising candidate in the field of cognitive enhancement and neuroprotection.

Suggested Dosing Information

200mcg - 500mcg prior to bed nightly

Benefits of Pinealon

1. Supports Cognitive Function: Pinealon helps improve brain function by promoting better communication between neurons, which enhances mental processing and cognitive abilities.
2. Improves Memory: Pinealon has been shown to improve both short-term and long-term memory by supporting brain cell health and improving neural connections.
3. Reduces Cognitive Decline: By protecting neurons from oxidative stress and age-related damage, Pinealon can slow down cognitive decline, particularly in aging individuals.
4. Enhances Overall Mental Health: Pinealon contributes to better mental health by improving cognitive function, reducing mental fatigue, and supporting emotional well-being.
5. Increases Focus: Pinealon boosts concentration and focus, helping individuals stay mentally sharp and productive, especially in complex tasks or learning environments.
6. Supports Neuroprotection: Pinealon provides neuroprotective benefits by defending brain cells against oxidative damage, inflammation, and other harmful factors that can lead to cognitive decline.
7. Improves Learning Retention: Pinealon enhances the brain's ability to absorb and retain new information, making it beneficial for learning and academic or professional settings.

COGNITIVE MODULATOR



Lyophilized Powder
Nasal Spray
SQ Injectable

N-Acetyl Selank

Selank is a synthetic peptide derived from the naturally occurring tetrapeptide tuftsin, known for its anxiolytic (anxiety-reducing) and nootropic (cognitive-enhancing) effects. It works by modulating the activity of neurotransmitters such as serotonin, dopamine, and norepinephrine, which are involved in mood regulation and cognitive function. Selank is primarily used to reduce anxiety, enhance memory, and improve mental clarity. Its ability to produce calming effects without sedation, combined with its cognitive-enhancing properties, makes Selank a valuable tool in the treatment of anxiety and cognitive disorders.

Suggested Dosing Information

Intranasal: 500mcg daily
SQ: 100mcg-300mcg daily

Benefits of N-Acetyl Selank

1. Enhances Cognitive Function: Selank helps improve brain function by promoting better neurotransmitter regulation, leading to improved mental performance and sharper cognition.
2. Supports Anxiety Reduction: Selank has been shown to reduce anxiety by modulating the brain's response to stress, providing a calming effect without the sedative side effects of traditional anti-anxiety medications.
3. Improves Mood: By promoting the balance of key neurotransmitters, such as serotonin, Selank can help improve mood and emotional well-being, making it useful for individuals dealing with mild depression or mood swings.
4. Enhances Overall Mental Clarity: Selank promotes mental clarity, helping users think more clearly and process information more efficiently.
5. Assists in Reducing Stress: Selank reduces stress levels by modulating the body's stress response, helping individuals feel more relaxed and composed in high-pressure situations.
6. Supports Neuroprotection: Selank has neuroprotective properties that help protect the brain from oxidative stress and other forms of damage, which is important for long-term cognitive health.
7. Increases Attention: Selank improves attention span and focus, making it easier to stay engaged in tasks and enhancing productivity in work or study settings.
8. Improves Cognitive Performance: By enhancing cognitive function and reducing mental fatigue, Selank improves overall cognitive performance, especially in tasks requiring sustained attention and memory recall.

COGNITIVE MODULATOR



Lyophilized Powder
Nasal Spray
SQ Injectable

N-Acetyl Semax

Semax is a synthetic peptide derived from adrenocorticotrophic hormone (ACTH), known for its nootropic, neuroprotective, and neuroregenerative properties. It works by modulating the activity of neurotransmitters and enhancing the expression of neurotrophic factors like BDNF (brain-derived neurotrophic factor). Semax is used to improve cognitive function, enhance memory, and protect against neurodegenerative diseases. It is also noted for its ability to reduce symptoms of stress and improve mood. Semax's multifaceted effects on brain health and cognitive function make it a promising candidate in the fields of nootropics and neuroprotection.

Suggested Dosing Information

Intranasal: 500mcg daily
SQ: 100mcg-300mcg daily

Benefits of N-Acetyl Semax

- 1. Supports Cognitive Enhancement:** Semax helps improve cognitive abilities by enhancing brain function, leading to better problem-solving, reasoning, and overall mental clarity.
- 2. Aids in Reducing Anxiety:** Semax has anxiolytic properties, meaning it can help reduce symptoms of anxiety without causing sedation, helping individuals remain calm and focused.
- 3. Assists in Improving Memory:** Semax enhances memory formation and recall by improving neural communication and boosting neurotransmitter activity, making it beneficial for learning and retention.
- 4. Enhances Overall Mental Function:** By supporting neurotransmitter systems and improving brain health, Semax leads to enhanced mental function, including faster processing speeds and sharper focus.
- 5. Increases Focus:** Semax improves attention and concentration, making it easier to stay on task and boosting productivity in work or academic settings.
- 6. Supports Neuroprotection:** Semax provides neuroprotective benefits by protecting brain cells from oxidative stress, inflammation, and neurodegenerative processes, promoting long-term brain health.
- 7. Assists in Managing Cognitive Disorders:** Semax shows potential in managing cognitive disorders such as ADHD, Alzheimer's, and other neurodegenerative conditions by supporting cognitive function and protecting neurons.
- 8. Improves Overall Brain Health:** By promoting blood flow to the brain, increasing neuroplasticity, and protecting against neural damage, Semax helps maintain and improve overall brain health and resilience.

COGNITIVE MODULATOR



**Capsule
Nasal Spray
Topical Cream**

Dihexa

Dihexa, also known as N-hexanoic-Tyr-Ile-(6), is a peptide derived from angiotensin IV, known for its potent cognitive-enhancing effects. It works by binding to hepatocyte growth factor (HGF) receptors, promoting synaptogenesis (the formation of new synapses) and enhancing neural plasticity. Dihexa is primarily researched for its potential in treating neurodegenerative diseases like Alzheimer's disease and for cognitive enhancement in healthy individuals. Its ability to cross the blood-brain barrier and directly stimulate brain cell growth makes Dihexa a promising candidate for improving memory, learning, and overall cognitive function.

Suggested Dosing Information

Oral - 5mg to 40mg daily
Nasal - 1.25mg to 2.5mg daily

Benefits of Dihexa

- 1.Cognitive Enhancement: Dihexa has shown potential in enhancing memory and learning capabilities, making it a candidate for treating cognitive impairments associated with brain injuries and neurodegenerative diseases.
- 2.Neuroprotective Effects: It promotes the growth and formation of synapses in the brain, which are essential for efficient neural communication. This synaptic enhancement contributes to better memory formation and retrieval, as well as overall brain health.
- 3.Antidementia Properties: Dihexa's potential to potentiate hepatocyte growth factor (HGF) and its receptor, c-Met, suggests that it could counteract the synaptic degradation seen in dementia, potentially slowing or reversing cognitive decline.
- 4.Manage Depression: By improving neural connectivity and potentially enhancing mood-regulating neurotransmitter functions, Dihexa may offer new avenues for managing symptoms of depression.
- 5.Increases Social Intuition: Improved cognitive processing could enhance the ability to read and respond to social cues, potentially increasing empathy and social intuition.
- 6.Improves Memory & Creative Thinking: Dihexa facilitates the growth and formation of synapses, crucial for memory retention and creativity, allowing for more innovative thinking and better recall of information.
- 7.Improve Focus & Learning: With enhanced synaptic connectivity, Dihexa can help sharpen focus and accelerate learning processes, making it easier to acquire new knowledge and skills.



MICHIGAN CENTER FOR
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GI HEALTH AND TISSUE REPAIR

GI health and tissue repair drugs are essential for maintaining the integrity of the digestive system and promoting the healing of damaged tissues. GI health drugs, such as proton pump inhibitors, antacids, and probiotics, help treat conditions like acid reflux, ulcers, irritable bowel syndrome (IBS), and inflammatory bowel diseases (IBD). These drugs improve digestion, reduce inflammation, and alleviate discomfort, leading to better nutrient absorption and overall gut health.

For individuals with chronic gastrointestinal issues, these medications provide relief from pain, bloating, and irregular bowel movements, allowing them to enjoy a more comfortable and active lifestyle. For those recovering from wounds, fractures, or surgeries, these medications can significantly enhance physical function, reduce pain, and help restore mobility.

Together, GI health and tissue repair drugs contribute to a higher quality of life by addressing both internal and external health concerns. They alleviate pain, promote healing, and enable individuals to recover faster and maintain better health, allowing them to lead more active, comfortable, and fulfilling lives.

GI HEALTH AND TISSUE REPAIR



**Capsule
Lyophilized Powder
SQ Injectable Topical
Cream**

N-Acetyl BPC-157

BPC-157, also known as Body Protective Compound-157, is a synthetic peptide derived from a protective protein found in the gastric juice. It is renowned for its potent regenerative and healing properties, particularly in the gastrointestinal tract, tendons, and muscles. BPC-157 works by promoting angiogenesis (the formation of new blood vessels) and accelerating the repair of damaged tissues. Clinically, it is used for the treatment of inflammatory bowel disease, gastric ulcers, and sports-related injuries. Its ability to enhance tissue repair and reduce inflammation makes BPC-157 a valuable peptide in both therapeutic and regenerative medicine.

Suggested Dosing Information

Oral: 500mcg daily
SQ QD: 400mcg - 600mcg SQ
BID: 200mcg - 300mcg

Benefits of N-Acetyl BPC-157

1. Enhanced Healing of Various Tissues: BPC-157 is notable for its ability to accelerate the healing of a wide array of tissues, including tendons, muscles, nervous system, and even bones. This makes it particularly useful for athletes and individuals recovering from injuries.
2. Gut Health Improvement: Given its origins in gastric juices, BPC-157 is inherently equipped to help with gastrointestinal issues. It can protect and heal intestinal walls and treat stomach ulcers, showing effectiveness in controlling inflammatory conditions such as IBD (inflammatory bowel disease).
3. Anti-inflammatory Properties: BPC-157 acts as a powerful anti-inflammatory, which is essential for healing and tissue repair. By modulating inflammatory pathways, it can reduce swelling and redness associated with injuries.
4. Pain Reduction: The peptide has been observed to reduce pain in damaged areas, possibly due to its anti-inflammatory effects and its ability to influence neurotransmitter systems and reduce nerve pain.
5. Cardiovascular Health: BPC-157 may help in repairing and protecting vascular tissue, thus improving blood flow and potentially lowering the risk of cardiovascular diseases.
6. Influence on Angiogenesis: The peptide promotes angiogenesis, which is the development of new blood vessels. Enhanced angiogenesis facilitates better blood flow to tissues, which is crucial for healing and recovery.

GI HEALTH AND TISSUE REPAIR



Capsule

Larazotide

Larazotide (AT 1001) is a peptide that acts as a tight junction regulator, helping to maintain the integrity of the intestinal barrier. It works by modulating the tight junctions between intestinal cells, preventing the leakage of harmful substances from the gut into the bloodstream, which is a common issue in conditions like celiac disease. Larazotide is primarily being developed as a treatment for celiac disease, as it has shown promise in reducing symptoms and intestinal permeability in response to gluten exposure. Its targeted action on tight junctions makes Larazotide a novel approach to managing autoimmune and inflammatory gut disorders.

Suggested Dosing Information

500mg x Twice Daily

Benefits of Larazotide Orally

1. Prevention of Gluten-Induced Symptoms: The primary benefit of larazotide is its ability to reduce symptoms associated with gluten ingestion in people with celiac disease. By inhibiting the action of zonulin and thereby maintaining tight junction integrity, larazotide can prevent the inflammatory response triggered by gluten.
2. Reduction in Intestinal Permeability: Larazotide helps maintain the barrier function of the gut, which is essential in preventing the translocation of harmful substances into the bloodstream. This reduction in intestinal permeability can help mitigate systemic inflammatory responses and may benefit other conditions characterized by increased gut permeability.
3. Potential Improvement in Gastrointestinal Health: By stabilizing the gut barrier, larazotide may help reduce symptoms of gastrointestinal discomfort, such as bloating, pain, and irregular bowel movements, particularly in conditions like celiac disease and potentially in other inflammatory bowel diseases.
4. Adjunct to Gluten-Free Diet: For individuals with celiac disease, even minute amounts of gluten can trigger significant health problems. Larazotide could potentially be used as an adjunct therapy to a gluten-free diet, providing an additional layer of protection against accidental gluten exposure.
5. Enhancing Quality of Life: The potential for larazotide to reduce the severity of symptoms associated with gluten exposure could significantly improve the quality of life for individuals with celiac disease, allowing them more dietary flexibility and reducing anxiety around food choices.



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GLP AND WEIGHT LOSS

GLP-1 (glucagon-like peptide-1) drugs are a type of weight loss medication that work by mimicking the natural GLP-1 hormone, which plays a key role in regulating appetite, blood sugar levels, and digestion. These drugs, originally developed to treat type 2 diabetes, have shown significant benefits for weight loss by helping individuals feel fuller for longer, reducing cravings, and slowing the emptying of the stomach, which can reduce calorie intake.

In addition to improving physical health, these drugs also enhance mental and emotional well-being. Weight loss can boost self-esteem, improve body image, and reduce anxiety or depression associated with obesity. Moreover, individuals often experience increased energy levels, better mobility, and a more active lifestyle, all of which contribute to a higher quality of life.

Overall, GLP-1 and other weight loss drugs offer a powerful tool for achieving and maintaining a healthier weight, leading to improved physical health, enhanced confidence, and greater overall life satisfaction.

SEMAGLUTIDE



**Lyophilized Powder
SQ Injectable**

Semaglutide

Semaglutide is a medication that belongs to the class of drugs known as glucagon-like peptide-1 (GLP-1) receptor agonists. It is a synthetic peptide that mimics the action of the naturally occurring hormone GLP-1, which is involved in the regulation of glucose metabolism. Semaglutide was developed for the treatment of type 2 diabetes and is also used for weight management in individuals with or without diabetes.

Semaglutide is a long-acting GLP-1 receptor agonist that stimulates insulin secretion and inhibits glucagon secretion in a glucose-dependent manner. Its chemical structure is similar to the human GLP-1 hormone but has been modified to provide a longer duration of action, allowing for once-weekly dosing.

Suggested Dosing Information

Week 1-4: 0.5mg weekly
Week 5-8: 1mg weekly
Week 9-12: 1.5mg weekly
Week 13-16: 2.0mg weekly
Week 17-20: 2.5mg weekly
Week 21+: 2.7mg weekly

Benefits of Semaglutide Injections

- 1. Improved Glycemic Control:** Semaglutide effectively lowers blood sugar levels in patients with type 2 diabetes by enhancing the secretion of insulin and suppressing the release of glucagon when glucose levels are high. It also slows gastric emptying, which helps in moderating blood glucose spikes after meals.
- 2. Weight Loss:** Semaglutide has a significant effect on weight reduction, making it a valuable option for managing obesity. It reduces appetite and food intake by acting on receptors in the brain that control hunger and satiety.
- 3. Cardiovascular Benefits:** Clinical trials have shown that Semaglutide can reduce the risk of major adverse cardiovascular events, such as heart attack, stroke, and cardiovascular death, in people with type 2 diabetes at high risk of such events.
- 4. Convenience of Use:** With its once-weekly injection schedule, Semaglutide offers a convenient treatment regimen, improving adherence compared to medications that require daily dosing.
- 5. Broad Therapeutic Potential:** Beyond diabetes and obesity, Semaglutide is being studied for potential effects in other metabolic, cardiovascular, and neurodegenerative disorders, indicating its broad therapeutic potential.



TIRZEPATIDE

**Lyophilized Powder
SQ Injectable**

Tirzepatide

Tirzepatide is a dual glucose-dependent insulintropic polypeptide (GIP) and glucagon-like peptide-1 (GLP-1) receptor agonist, used for the treatment of type 2 diabetes. It is designed to improve glycemic control, body weight, and other metabolic parameters in people with type 2 diabetes. The combination of GIP and GLP-1 receptor agonism in tirzepatide is intended to provide complementary mechanisms of action to address multiple facets of type 2 diabetes pathophysiology, including insulin secretion, insulin sensitivity, and appetite regulation.

Tirzepatide has been studied in clinical trials, and results have shown promising efficacy in reducing HbA1c (a measure of long-term blood sugar control) and body weight compared to other antidiabetic medications.

Suggested Dosing Information

Week 1-4: 2.5mg weekly
Week 5-8: 5mg weekly
Week 9-12: 7.5mg weekly
Week 13-16: 10mg weekly
Week 17-20: 12.5mg weekly
Week 21+ 15mg weekly

Benefits of Tirzepatide Injections

- 1.Enhanced Glycemic Control: Tirzepatide has shown superior efficacy in lowering blood sugar levels compared to other GLP-1 receptor agonists. It improves insulin secretion in a glucose-dependent manner, reduces glucagon secretion, and delays gastric emptying, all of which contribute to improved glycemic control in patients with type 2 diabetes.
- 2.Significant Weight Loss: Clinical trials have demonstrated that tirzepatide can lead to substantial weight loss, often surpassing results seen with other diabetes medications. This effect is due to its dual action on GIP and GLP-1 receptors, which not only helps to control blood sugar but also suppresses appetite and increases energy expenditure.
- 3.Potential Cardiovascular Benefits: While the specific cardiovascular effects are still being studied, the weight loss and improved metabolic control associated with tirzepatide could potentially reduce cardiovascular risk factors in patients with type 2 diabetes.
- 4.Convenience of Use: Tirzepatide is administered via subcutaneous injection once a week, which can be more convenient for patients compared to medications requiring daily dosing, thereby improving adherence to the treatment regimen.
- 5.Potential in Treating Obesity in Non-Diabetics: Given its significant impact on weight reduction, tirzepatide is also being explored as a treatment for obesity in individuals without diabetes. This could broaden its application as a therapeutic agent in a wider population struggling with weight management.



MICHIGAN CENTER FOR
REGENERATIVE MEDICINE



IMMUNO- MODULATING

Immuno-modulating drugs are medications designed to adjust the immune system's activity, either enhancing or suppressing its function, depending on the condition being treated. These drugs are essential in managing autoimmune diseases, inflammatory disorders, cancers, and conditions that involve an overactive or weakened immune response. By regulating immune function, immuno-modulating drugs help restore balance in the body's defense mechanisms, promoting overall health and preventing disease progression.

On the other hand, in patients with weakened immune systems, such as those undergoing cancer treatment or with chronic infections, immuno-modulating drugs can boost immune responses, helping the body fight off infections and recover more effectively.

By reducing the severity of symptoms and improving the body's ability to fight disease, immuno-modulating drugs significantly enhance physical function, reduce pain, and improve energy levels, enabling individuals to live more comfortable and active lives.

IMMUNO-MODULATING



THYMOSIN BETA-4

Lyophilized Powder
SQ Injectable
Nasal Spray

Thymosin Beta-4

Thymosin Beta-4 is a naturally occurring peptide that plays a crucial role in tissue repair and regeneration. It works by promoting cell migration, angiogenesis (formation of new blood vessels), and wound healing. Thymosin Beta-4 is primarily used for its regenerative properties in treating injuries, chronic wounds, and heart conditions such as myocardial infarction. Its ability to enhance tissue repair and reduce inflammation makes Thymosin Beta-4 a valuable therapeutic agent in regenerative medicine and tissue healing.

Suggested Dosing Information

300mcg - 1g daily QD or BID

Benefits of Thymosin Beta-4

1. Supports Tissue Repair: Thymosin Beta-4 promotes the repair of damaged tissues by facilitating cell migration and regeneration, making it useful in injury recovery.
2. Aids in Reducing Inflammation: TB-500 helps reduce inflammation, alleviating swelling and pain, which can speed up recovery from injuries or surgeries.
3. Improves Wound Healing: It accelerates the healing process of wounds by promoting new blood vessel formation (angiogenesis) and collagen production, which are essential for tissue repair.
4. Enhances Overall Tissue Regeneration: Thymosin Beta-4 supports the regeneration of tissues such as muscle, skin, and connective tissue, aiding in overall body recovery and repair.
5. Reduces Chronic Pain: By reducing inflammation and enhancing tissue repair, TB-500 may help alleviate chronic pain, particularly in conditions related to joint, muscle, or soft tissue damage.
6. Supports Muscle Recovery & Flexibility: It enhances muscle recovery and flexibility by promoting cell repair and reducing stiffness, making it useful for athletes or individuals recovering from muscle injuries.
7. Improves Overall Skin Health: Thymosin Beta-4 supports skin repair and regeneration, improving skin elasticity and overall appearance by enhancing the healing of cuts, scars, and other skin conditions.
8. Enhances Cellular Repair: TB-500 aids in cellular repair processes, particularly in tissues damaged by injury, oxidative stress, or inflammation, supporting long-term tissue health.

IMMUNO-MODULATING



**Lyophilized Powder
SQ Injectable**

Thymulin

Thymulin is a thymic peptide that plays a critical role in modulating the immune system. It works by enhancing the activity of T-cells, which are essential for immune function. Thymulin is primarily used to support immune health, particularly in conditions where the immune system is compromised, such as autoimmune diseases, chronic infections, and aging. Its ability to boost immune function and regulate inflammatory responses makes Thymulin a promising candidate in the field of immunotherapy and immune modulation.

Suggested Dosing Information

500mcg - 1000mcg daily

Benefits of Thymulin Injections

1. Immune Regulation: Thymulin plays a significant role in regulating the immune system. Its presence helps to maintain and modulate the immune response, ensuring that it remains effective yet not excessively aggressive, which can be particularly beneficial in autoimmune diseases where the immune system attacks the body's own tissues.
2. Anti-inflammatory Effects: Thymulin has anti-inflammatory properties. It can modulate the body's inflammatory response, which is beneficial in treating chronic inflammatory diseases and may help reduce the severity of conditions like arthritis and asthma.
3. Enhancement of T-Cell Function: By promoting the differentiation and function of T-cells, Thymulin injections can enhance cellular immunity. This is particularly important for individuals with weakened immune systems, such as the elderly or those undergoing treatments that suppress immune function like chemotherapy.
4. Potential Role in Allergy and Asthma: Thymulin may help modulate allergic responses and asthma. Its ability to regulate immune responses can potentially reduce the hypersensitivity reactions characteristic of these conditions.
5. Neuroendocrine Regulation: There is evidence suggesting that Thymulin may also play a role in the neuroendocrine system, influencing the interaction between the nervous and endocrine systems and affecting how the body responds to stress.

IMMUNO-MODULATING



Lyophilized Powder
SQ Injectable

Thymosin Alpha-1

Thymosin Alpha-1 is a 28-amino acid immunomodulatory peptide naturally produced by the thymus and classified as an immune system activator. It supports T-cell maturation and helps regulate immune balance by enhancing the function of natural killer cells, dendritic cells, and macrophages. The peptide acts on immune tissues such as the thymus, lymph nodes, and sites of infection or inflammation to strengthen the body's defense mechanisms. Its downstream effects include increased immune surveillance, better pathogen clearance, and more regulated inflammatory responses. When administered subcutaneously, it offers systemic distribution with effects that last several hours to days depending on the immune condition. Thymosin Alpha-1 primarily influences immune, infectious disease, and inflammatory pathways. It demonstrates selectivity for dysregulated or suppressed immune states without overstimulating healthy immune function.

Suggested Dosing Information

500-1500mcg daily

Benefits of Thymosin Alpha-1

1. Boosts Immune Function: Thymosin Alpha-1 enhances T-cell activity, helping the body fight infections more effectively.
2. Improves Vaccine Response: It has been shown to strengthen the immune system's response to vaccines, making them more effective.
3. Fights Viral Infections: Thymosin Alpha-1 supports the immune system in combating viruses such as hepatitis and influenza.
4. Supports Cancer Therapy: It may enhance the effectiveness of certain cancer treatments by improving immune recognition of tumor cells.
5. Reduces Inflammation: Thymosin Alpha-1 helps regulate the immune response to lower harmful chronic inflammation.
6. Protects Against Immune Suppression: It can help restore immune balance in patients with weakened or suppressed immune systems.
7. Promotes Healthy Aging: By maintaining immune system strength, Thymosin Alpha-1 supports resilience and overall well-being with age.

IMMUNO-MODULATING

N-ACETYL KPV AMIDATE

Capsule
SQ Injectable
Lyophilized Powder
Topical Cream

N-Acetyl KPV Amidate

KPV is a tripeptide with potent anti-inflammatory and immunomodulatory properties. It is derived from the naturally occurring alpha-melanocyte-stimulating hormone (α -MSH) and is known for its ability to reduce inflammation and promote healing, particularly in conditions such as inflammatory bowel disease (IBD) and wound healing. KPV works by inhibiting pro-inflammatory cytokines and reducing oxidative stress, making it effective in treating chronic inflammatory conditions. Its broad-spectrum anti-inflammatory effects and potential in promoting tissue repair make KPV a promising therapeutic agent in both autoimmune and inflammatory diseases.

Suggested Dosing Information

250mcg - 500mcg SQ daily

Benefits of N-Acetyl KPV Amidate

Injections

1. Direct and Potent Effects: KPV injections can deliver the peptide directly into the bloodstream, ensuring immediate and potent therapeutic effects.
2. Rapid Response: Injections are useful for rapid relief from acute inflammatory conditions or flare-ups in chronic diseases.
3. Specific Targeting: Injectable KPV can be used for localized treatment, such as in joint inflammation or specific sites of autoimmune activity.

Capsules

1. Convenience: Oral capsules offer a convenient method of delivery, especially for long-term treatment.
2. Systemic Distribution: When formulated for proper absorption, capsules can provide a systemic distribution of KPV, potentially benefiting conditions that affect multiple body systems.
3. Sustained Release: Capsules can be designed to release KPV slowly over time, maintaining a steady level of the peptide in the bloodstream.

Creams

1. Targeted Skin Treatment: Topical application of KPV in the form of creams is ideal for treating skin conditions such as eczema, psoriasis, and other inflammatory dermatoses.
2. Localized Effect with Minimal Systemic Absorption: Creams allow for high local concentrations of KPV without significant systemic effects, reducing the risk of side effects.
3. Ease of Use: KPV creams can be applied directly to the affected areas, making this form user-friendly and non-invasive.

IMMUNO-MODULATING



N-Acetyl Epitalon

Epitalon, also known as epithalon or Epithalamin, is a synthetic peptide known for its role in regulating the body's circadian rhythms and enhancing the activity of telomerase, an enzyme that protects DNA and promotes cellular longevity. It is derived from the naturally occurring pineal gland peptide epithalamin. Epitalon is primarily used for its anti-aging effects, including the potential to extend lifespan, improve sleep quality, and enhance immune function. By promoting telomerase activity, Epitalon helps maintain the integrity of chromosomes, thereby slowing down the aging process at the cellular level. Its broad effects on aging and health make it a cornerstone in anti-aging research and therapy.

Suggested Dosing Information

10mg SQ q3D x 5 doses

2mg SQ q3D x 10 doses

Benefits of N-Acetyl Epitalon

1. Supports Anti-Aging Effects: Epitalon is believed to slow down the aging process by enhancing telomerase activity, which helps maintain telomere length, thereby protecting cells from aging and degradation.
2. Aids in Extending Lifespan: By improving cellular repair mechanisms and reducing age-related damage, Epitalon may help extend the lifespan of cells and, potentially, the individual.
3. Reduces Oxidative Stress: Epitalon has antioxidant properties, helping to neutralize free radicals that cause oxidative stress, which is a major contributor to aging and chronic diseases.
4. Enhances Cellular Repair: By promoting telomerase production, Epitalon aids in the repair and regeneration of damaged cells, which is crucial for maintaining youthful and healthy tissues.
5. Improves Overall Health: Epitalon may enhance general health by regulating the production of key hormones such as melatonin, improving sleep, and supporting the immune system.
6. Supports Longevity: Epitalon's ability to promote telomerase activity, reduce oxidative damage, and enhance DNA repair mechanisms supports long-term health and longevity.
7. Manages Age-Related Disorders: By supporting cellular function and reducing oxidative stress, Epitalon may help manage or prevent age-related disorders like cardiovascular diseases, neurodegenerative conditions, and certain types of cancer.



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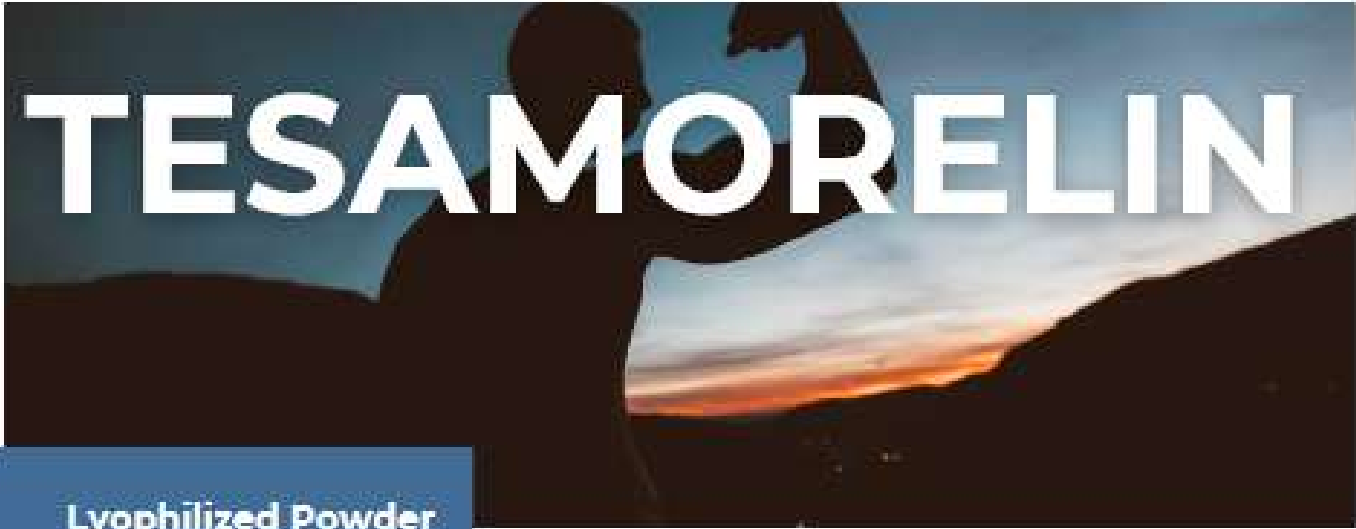


LONGEVITY

Longevity drugs are designed to stimulate the release of growth hormone from the pituitary gland, which plays a vital role in regulating metabolism, muscle growth, and tissue repair. These drugs mimic the body's natural mechanisms to increase growth hormone production, which tends to decline with age. By doing so, longevity drugs offer a range of health benefits that can significantly improve quality of life, particularly for those experiencing age-related physical decline.

One of the primary benefits of longevity drugs is their ability to enhance muscle mass and strength, making them valuable for individuals suffering from sarcopenia (age-related muscle loss) or those recovering from injuries. Increased muscle mass and better physical strength translate into improved mobility, balance, and overall independence, especially as people age.

Overall, longevity drugs offer a powerful tool for maintaining physical health, energy, and resilience into later years, contributing to a more vibrant, active, and fulfilling life.



TESAMORELIN

Lyophilized Powder
SQ Injectable

Tesamorelin

Tesamorelin, also known as growth hormone-releasing factor (GRF) or TH9507, is a synthetic analog of growth hormone-releasing hormone (GHRH) that is used to stimulate the release of growth hormone (GH) from the pituitary gland. It is primarily used to reduce visceral fat in HIV-infected patients with lipodystrophy, a condition characterized by abnormal fat distribution. Tesamorelin works by increasing GH levels, which in turn promotes fat loss, improves body composition, and enhances overall metabolic health. Its targeted action on GH release and effectiveness in reducing visceral fat make Tesamorelin a valuable therapeutic agent in both clinical and metabolic health settings.

Suggested Dosing Information

500mcg - 1000mcg 90 minutes away from food daily

Benefits of Tesamorelin

- 1.Reduction in Visceral Abdominal Fat: The primary clinical benefit of Tesamorelin is its ability to significantly reduce visceral fat in the abdominal area without affecting subcutaneous fat. This is particularly beneficial for HIV patients experiencing lipodystrophy, as it can help manage the increased risk of cardiovascular disease associated with excess visceral fat.
- 2.Improved Lipid Profiles: Tesamorelin has been shown to improve lipid levels, decreasing total cholesterol and LDL cholesterol, while potentially increasing HDL cholesterol. This can contribute to a lower risk of cardiovascular diseases.
- 3.Possible Cognitive Benefits: Some studies suggest that Tesamorelin might improve cognitive function, particularly in older adults or those with cognitive impairment. This is thought to be related to its effects on growth hormone levels, which can influence brain health.
- 4.Increased Muscle Mass: As a growth hormone-releasing hormone, Tesamorelin can help increase lean muscle mass, which in turn may improve physical strength and mobility.



HEXARELIN

Capsule

Hexarelin

Hexarelin, also known as examorelin, is a synthetic hexapeptide and potent growth hormone secretagogue, known for its ability to significantly increase the release of growth hormone (GH) from the pituitary gland. It works by binding to the growth hormone secretagogue receptor (GHSR), stimulating GH release, which in turn promotes muscle growth, enhances fat metabolism, and improves recovery. Hexarelin is also noted for its cardioprotective effects, making it a potential treatment for heart disease. Its strong GH-releasing capabilities, along with its benefits for muscle and heart health, make Hexarelin a promising peptide in both clinical and performance-enhancement contexts.

Suggested Dosing Information

Oral: 10mg to 20mg fasted in the evening daily

SQ: 100mcg to 300mcg fasted in the evening daily

Nasal: 1.25mg fasted in the evening daily

Benefits of Hexarelin Injections

1. Increased Growth Hormone Levels: The primary benefit of Hexarelin injections is their ability to significantly boost growth hormone levels. Increased GH levels can have various positive effects on the body, such as improved muscle growth and strength, enhanced metabolism, and better overall cellular repair and regeneration.
2. Enhanced Muscle Growth and Strength: By increasing growth hormone levels, Hexarelin can help enhance muscle mass and strength. This makes it particularly appealing for athletes and bodybuilders looking to improve their performance and physique.
3. Improved Fat Metabolism: Hexarelin may assist in reducing body fat through its impact on growth hormone levels. GH is known to enhance lipolysis, the process of breaking down fat cells, which can lead to reduced body fat percentage.
4. Anti-Aging Effects: Growth hormone is often noted for its anti-aging capabilities. Hexarelin's ability to increase GH levels may help improve skin elasticity, reduce wrinkles, and support healthier hair and nails.
5. Cardioprotective Properties: Some studies suggest that Hexarelin may have cardioprotective effects. It can potentially improve heart function, particularly after cardiac events like heart attacks, by promoting cardiac cell repair and regeneration.
6. Enhanced Recovery and Healing: Due to its impact on growth hormone levels, Hexarelin may accelerate the healing process after injuries or surgeries, particularly in terms of tissue regeneration and recovery time.



N-ACETYL IPAMORELIN

Lyophilized Powder
SQ Injectable

N-Acetyl Ipamorelin

Ipamorelin is a synthetic pentapeptide that functions as a growth hormone secretagogue, specifically targeting the ghrelin receptor to stimulate the pituitary gland's release of growth hormone (GH). Unlike some other GH secretagogues, Ipamorelin exhibits a high degree of selectivity, minimizing the activation of cortisol and prolactin, which can lead to fewer side effects. Clinically, Ipamorelin is often utilized for its potential to enhance muscle hypertrophy, improve recovery from injuries, and support lipolysis, making it beneficial in both therapeutic and performance enhancement contexts. Additionally, it is explored in anti-aging medicine due to its ability to promote collagen synthesis, enhance skin elasticity, and improve overall vitality.

Suggested Dosing Information

100mcg - 400mcg SQ in the evening 90 minutes away from food 5 days per week

Benefits of N-Acetyl Ipamorelin

1. Increased Growth Hormone Levels: Ipamorelin effectively increases the production and release of growth hormone in the body, which has a variety of positive effects such as enhanced tissue growth and regeneration.
2. Fat Loss and Lean Muscle Gain: By stimulating growth hormone release, Ipamorelin helps to enhance the body's ability to burn fat and build lean muscle mass, making it a popular choice among athletes and bodybuilders for its body composition benefits.
3. Anti-Aging Effects: Higher levels of growth hormone can contribute to improved skin elasticity, reduced wrinkles, and other signs of aging. Growth hormone also promotes the regeneration of cells, which is essential for maintaining youthful tissue function.
4. Improved Recovery and Healing: The increased growth hormone levels promoted by Ipamorelin enhance the body's ability to repair itself, which is beneficial for recovery from injuries and intense physical activities. This can lead to quicker recovery times and improved overall resilience.
5. Better Sleep Quality: Ipamorelin has been shown to improve sleep patterns. Growth hormone itself plays a role in the quality and depth of sleep, which is critical for optimal health and well-being.
6. Enhanced Bone Density: Growth hormone stimulates bone growth and maintenance, which can lead to increased bone density over time. This is particularly beneficial in aging populations who are at risk of osteoporosis.



SERMORELIN

Lyophilized Powder
SQ Injectable
Troche

Sermorelin

Sermorelin is a synthetic peptide analog of growth hormone-releasing hormone (GHRH), used to stimulate the production and release of growth hormone (GH) from the pituitary gland. It works by mimicking the natural GHRH, thereby enhancing GH secretion, which promotes muscle growth, fat loss, and overall vitality. Sermorelin is primarily used in anti-aging therapy to improve energy levels, enhance muscle mass, and support healthy aging. Its ability to naturally stimulate GH production makes Sermorelin a popular choice in both therapeutic and performance-enhancing settings.

Suggested Dosing Information

200mcg - 300mcg 90 minutes away from
food AM/PM daily

Benefits of Sermorelin

1. Stimulates Natural Growth Hormone Production: Sermorelin works by directly stimulating the pituitary gland to produce more growth hormone. This is a more natural approach compared to direct GH injections, as it encourages the body to regulate and secrete GH as needed.
2. Improves Body Composition: Increased levels of GH can lead to a reduction in body fat and an increase in lean muscle mass. This makes Sermorelin a popular choice not only for individuals with GH deficiency but also for those looking to enhance body composition as part of an anti-aging protocol or health regimen.
3. Anti-Aging Effects: By restoring growth hormone levels to a more youthful state, Sermorelin can help improve skin elasticity, reduce wrinkles, and promote a healthier overall appearance.
4. Enhances Physical Performance and Recovery: Increased growth hormone levels can improve physical performance, stamina, and recovery times from injuries. This is particularly beneficial for athletes or individuals engaged in regular physical activity.
5. Improves Sleep Patterns: Growth hormone plays a significant role in sleep regulation, particularly in promoting deep, restorative sleep. Sermorelin can help enhance the quality of sleep, which is crucial for overall health and well-being.
6. Bone Health: GH is vital for maintaining and improving bone density. Sermorelin therapy can help prevent osteoporosis and strengthen bone health, particularly important as one ages.



MICHIGAN CENTER FOR
REGENERATIVE MEDICINE



MITOCHONDRIAL MIMETIC

Mitochondrial mimetics are a class of drugs and compounds designed to replicate or enhance the functions of mitochondria, the energy-producing organelles in cells. These compounds aim to boost mitochondrial efficiency, reduce oxidative stress, and improve cellular energy production. Mitochondrial mimetics are particularly valuable for counteracting the effects of aging and treating conditions associated with mitochondrial dysfunction, such as neurodegenerative diseases, metabolic disorders, and fatigue syndromes.

One of the key benefits of mitochondrial mimetics is their ability to increase energy levels by optimizing the production of ATP (adenosine triphosphate), the energy currency of cells. This can lead to reduced fatigue and improved endurance, allowing individuals to remain more active and productive throughout the day.

Overall, mitochondrial mimetics offer a promising approach to enhancing energy, protecting against disease, and promoting healthy aging, contributing to a more active, resilient, and fulfilling life.

MITOCHONDRIAL MIMETIC



Capsule
Lyophilized Powder
SQ Injectable

MOTs-C

MOTS-C, which stands for Mitochondrial Open Reading Frame of the 12S rRNA-C, is a mitochondrial-derived peptide that plays a significant role in regulating metabolism and insulin sensitivity. It works by activating the AMPK pathway, which is crucial for energy balance and metabolic homeostasis. MOTs-C has been shown to enhance glucose uptake, improve insulin sensitivity, and promote fat loss, making it a promising candidate for treating metabolic disorders such as obesity and type 2 diabetes. Additionally, MOTs-C is being explored for its potential to improve exercise performance and increase longevity. Its ability to modulate metabolic function at the cellular level makes MOTs-C a key player in metabolic health research.

Suggested Dosing Information

MOTs-C analog oral: 2.5mg to 5mg daily
N-Acetyl MOTs-C SQ: 5mg daily

Benefits of MOTs-C analog Injections

1. Supports Mitochondrial Function: MOTS-c enhances the efficiency and performance of mitochondria, the energy centers of cells, ensuring they produce optimal levels of energy to support bodily functions.
2. Improves Cellular Energy: By boosting mitochondrial activity, MOTS-c helps increase cellular energy production, reducing fatigue and promoting better physical and mental performance throughout the day.
3. Reduces Oxidative Stress: MOTS-c plays a role in combating oxidative stress by neutralizing free radicals, protecting cells from damage, and slowing the aging process, which contributes to better long-term health.
4. Enhances Overall Vitality: With improved cellular function and reduced stress on the body, MOTS-c helps increase vitality, making individuals feel more energetic, resilient, and vibrant.
5. Improves Metabolic Function: MOTS-c promotes healthy metabolism by enhancing insulin sensitivity and regulating blood sugar levels, which is beneficial for individuals looking to manage weight and prevent metabolic diseases like type 2 diabetes.
6. Supports Cognitive Health: MOTS-c may protect brain cells from oxidative damage and improve energy supply to the brain, leading to enhanced cognitive function, better focus, and memory retention.
7. Increases Endurance: By improving mitochondrial function and energy production, MOTS-c helps boost physical endurance, allowing individuals to perform physical activities for longer periods without fatigue.
8. Supports Athletic Performance: Athletes and active individuals may benefit from MOTS-c's ability to enhance muscle function, increase stamina, and promote faster recovery, making it an effective tool for improving overall athletic performance.

MITOCHONDRIAL MIMETIC



**Lyophilized Powder
SQ Injectable**

Humanin

Humanin is a mitochondrial-derived peptide that has garnered attention for its neuroprotective and anti-apoptotic (cell death-preventing) properties. It acts by interacting with several cell surface receptors to inhibit cell death pathways, making it a potential therapeutic agent for neurodegenerative diseases such as Alzheimer's and Parkinson's disease. Humanin also has cardioprotective effects, helping to reduce the damage caused by ischemia-reperfusion injury in the heart. Its role in modulating mitochondrial function and preventing cellular apoptosis makes Humanin a promising peptide for improving longevity and protecting against age-related diseases.

Suggested Dosing Information

1mg daily

Benefits of Humanin Peptide Injections

- 1. Neuroprotective Effects:** Humanin has shown promising results in neuroprotection, particularly in models of Alzheimer's disease. It seems to protect neurons from amyloid-beta-induced toxicity, which is a hallmark of Alzheimer's pathology.
- 2. Mitochondrial Protection:** Humanin also helps in protecting mitochondria, the powerhouses of cells, from damage. This mitochondrial protective effect is crucial in preventing cell death and enhancing cell function across various tissues.
- 3. Metabolic Regulation:** There is evidence suggesting that Humanin can influence metabolic processes, including insulin sensitivity and glucose metabolism. This could be beneficial for conditions like diabetes and metabolic syndrome.
- 4. Cardioprotective Properties:** Humanin has been studied for its potential to protect against heart disease. It may help reduce the impact of oxidative stress and inflammation in cardiac cells, potentially reducing the risk of heart failure and other cardiovascular diseases.
- 5. Anti-Apoptotic Effects:** By preventing apoptosis (programmed cell death), Humanin can help maintain cell survival in response to stress or disease conditions. This is beneficial in a variety of pathological conditions where cell death is a primary concern.
- 6. Potential Anti-Aging Effects:** Preliminary studies suggest that Humanin might play a role in aging processes, potentially extending lifespan or improving the health span by protecting cells and tissues from age-related decline.

MITOCHONDRIAL MIMETIC



Capsule

Methylene Blue

Methylene Blue is a redox-active compound classified as a mitochondrial enhancer and neuroprotective dye. It facilitates more efficient electron transfer within the mitochondrial respiratory chain, improving cellular energy production and reducing oxidative damage. Methylene Blue targets high-energy-demand tissues such as the brain, heart, and muscles, where it supports ATP output and mitochondrial repair. Downstream effects include improved mental stamina, protection against neurodegeneration, and greater cellular resilience under stress. When taken orally, it offers sustained absorption and extended bioavailability. It primarily supports the neurologic and metabolic systems without altering hormone levels or neurotransmitter activity. Tissue effects are focused on mitochondria-rich cells, making it valuable in fatigue, brain fog, and aging protocols. Its unique mechanism complements other metabolic therapies in performance and recovery plans.

Suggested Dosing Information

25mg - 100mg daily

Benefits of Methylene Blue

1. Enhances Mitochondrial Function: Methylene Blue acts as an alternative electron carrier, improving cellular energy production and efficiency.
2. Supports Cognitive Performance: It enhances memory, focus, and mental clarity by protecting and energizing neurons.
3. Provides Neuroprotection: By reducing oxidative stress and supporting brain metabolism, it may help protect against neurodegenerative conditions.
4. Acts as a Potent Antioxidant: Methylene Blue neutralizes free radicals, reducing cellular damage from oxidative stress.
5. Improves Cellular Resilience: It supports cell survival and function under stress, enhancing overall cellular health.
6. Promotes Longevity Pathways: By improving mitochondrial function and reducing oxidative damage, it may contribute to healthy aging.
7. Supports Mood and Mental Health: Low doses of Methylene Blue have shown potential in improving mood, reducing anxiety, and supporting overall emotional well-being.



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REPAIR AND RECOVERY

Repair and recovery drugs are designed to accelerate the healing process, enhance tissue regeneration, and reduce inflammation, aiding in the recovery from injuries, surgeries, and chronic conditions. These drugs play a crucial role in supporting the body's natural repair mechanisms, promoting faster recovery and improving overall well-being. Whether it's for wound healing, muscle repair, or managing conditions like arthritis, these medications help restore normal function more quickly and effectively.

One of the main benefits of repair and recovery drugs is their ability to reduce pain and inflammation, allowing individuals to heal with less discomfort. These drugs also promote faster tissue regeneration, helping to rebuild damaged muscles, bones, skin, or other tissues. Additionally, by supporting quicker recovery, these drugs help individuals return to their normal activities faster, improving their emotional and psychological well-being.

In summary, repair and recovery drugs offer significant benefits by reducing pain, accelerating healing, and restoring physical function, allowing individuals to recover faster, stay active, and enjoy a higher quality of life.

REPAIR AND RECOVERY



**Nasal Spray
Topical Cream
Topical Solution**

GHK-Cu

GHK-Cu (Glycyl-L-histidyl-L-lysine-Copper) is a naturally occurring copper peptide known for its regenerative and anti-aging properties. It functions by activating a wide range of processes, including wound healing, collagen production, and antioxidant defense. GHK-Cu is also involved in modulating gene expression, particularly genes related to inflammation, tissue repair, and skin health. Clinically, it is used in skin care for its ability to improve skin elasticity, reduce wrinkles, and promote healing. Additionally, GHK-Cu is being researched for its potential to treat conditions like COPD, cancer, and neurodegenerative diseases due to its ability to repair and regenerate damaged tissues. GHK-Cu is a combination of the amino acids glycine (G), histidine (H), and lysine (K), with a copper ion (Cu) attached.

Suggested Dosing Information

Nasal: 1.25mg-2.5mg daily

Topical: 2.5mg daily

Benefits of GHK-Cu

1. Promotes Wound Healing: GHK-Cu accelerates wound healing by promoting the migration of repair cells to the injury site and stimulating tissue repair.
2. Supports Skin Repair: It enhances the skin's ability to repair itself from damage caused by injuries, environmental factors, or aging.
3. Reduces Signs of Aging: GHK-Cu helps reduce wrinkles, fine lines, and other signs of aging by stimulating collagen production and promoting skin rejuvenation.
4. Enhances Tissue Regeneration: The peptide promotes the regeneration of tissues, making it useful for healing wounds, repairing damaged skin, and rejuvenating cells.
5. Improves Skin Elasticity: By boosting collagen and elastin production, GHK-Cu helps improve skin elasticity, leading to firmer and more youthful-looking skin.
6. Supports Collagen Production: One of its key benefits is the stimulation of collagen synthesis, which plays a vital role in maintaining healthy, plump skin.
7. Reduces Inflammation: GHK-Cu has anti-inflammatory properties that help reduce redness, swelling, and irritation, promoting a calmer, clearer complexion.
8. Improves Overall Skin Health: Its ability to repair skin, reduce oxidative damage, and support cellular regeneration leads to improved skin texture, tone, and hydration.
9. Aids in Hair Regrowth: GHK-Cu has shown potential in stimulating hair follicle growth, promoting hair regeneration and thickening, making it useful in combating hair loss.

REPAIR AND RECOVERY



GLUTATHIONE

Capsule
SQ Injectable
Nasal Spray

Inhalant
Cream

Glutathione

Glutathione is a tripeptide composed of glutamine, cysteine, and glycine, and is the body's most important intracellular antioxidant. It is found in every cell and is critical for detoxification, immune function, and protecting cells from oxidative stress. Glutathione neutralizes free radicals and reactive oxygen species, supports liver detoxification processes, and regenerates other antioxidants such as vitamin C and E. In IV therapy, glutathione is used to enhance detoxification, reduce oxidative stress, boost immune function, and improve overall health and vitality.

Suggested Dosing Information

Topical - 1000mcg daily
Oral - 125mg to 250mg daily
Injection - 100mg daily
Nasal - 15mg to 30mg daily
Inhale - 100mg as needed

Benefits of Glutathione

1. Antioxidant Protection: Glutathione directly scavenges reactive oxygen species (ROS) and helps to maintain other antioxidants in their active forms. It is vital for preventing cellular damage, aging, and various diseases, including cancer and heart disease.
2. Detoxification: The liver uses glutathione in a number of detoxification processes. It binds to toxins, heavy metals, and by-products of metabolism, making them more water-soluble and easier to excrete.
3. Immune System Support: Glutathione enhances the immune system's function by influencing lymphocytes, particularly increasing the production of T-cells, which are essential for adaptive immunity.
4. Skin Health: Adequate levels of glutathione in the body can contribute to a bright, healthy complexion. It is often used in cosmetic products for its skin-lightening effects because it inhibits the enzyme tyrosinase, which is involved in melanin production.
5. Anti-aging Properties: By reducing oxidative stress and improving detoxification, glutathione helps to slow down the aging process. It may help reduce age-related conditions like Alzheimer's disease, macular degeneration, and arthritis.
6. Increased Energy: Glutathione plays a significant role in the energy production processes of cells. By improving mitochondrial function, it can lead to increased energy and stamina.



SEXUAL FUNCTION

Sexual function drugs are medications designed to address sexual dysfunction in both men and women, improving various aspects of sexual health and enhancing quality of life. These drugs are commonly used to treat conditions such as erectile dysfunction (ED), low libido, and other forms of sexual performance issues.

The benefits of sexual function drugs extend beyond the physical aspects of sexual health. These medications can improve emotional well-being by restoring intimacy and strengthening personal relationships, which can reduce anxiety, boost self-esteem, and improve mental health.

Overall, sexual function drugs contribute significantly to improving sexual health, emotional satisfaction, and overall quality of life by addressing both the physical and psychological aspects of sexual dysfunction.

SEXUAL FUNCTION



Lyophilized Powder
Nasal Spray
SQ Injectable
Troche

PT-141 (Bremelanotide)

Bremelanotide, also known as PT-141, is a synthetic peptide known for its ability to enhance sexual arousal and treat sexual dysfunction. It works by activating melanocortin receptors in the brain, which are involved in sexual desire and arousal. PT-141 is primarily used to treat conditions like erectile dysfunction and female sexual arousal disorder. Unlike traditional erectile dysfunction medications, PT-141 acts on the central nervous system rather than directly on the vascular system, making it effective for both men and women. Its ability to enhance sexual desire and arousal makes PT-141 a unique and promising therapeutic agent in sexual medicine.

Suggested Dosing Information

SQ: 1mg SQ 60 minutes prior to sexual activity as needed.

Nasal: 1-2 sprays each nostril 60 minutes prior to sexual activity as needed.

Troche: 2-4mg prior to sexual activity as needed.

Benefits of PT-141 (Bremelanotide)

- 1. Enhances Sexual Function:** PT-141 stimulates sexual desire and arousal by activating specific receptors in the brain that control sexual function, improving overall sexual response.
- 2. Supports Libido:** It effectively increases libido in both men and women, making it beneficial for those experiencing low sexual desire.
- 3. Improves Erectile Function:** PT-141 has been shown to enhance erectile function by promoting blood flow, helping men achieve and maintain erections more effectively.
- 4. Assists in Sexual Dysfunction Treatment:** It is used to treat sexual dysfunction in both men and women, including conditions like erectile dysfunction (ED) in men and hypoactive sexual desire disorder (HSDD) in women.
- 5. Enhances Sexual Arousal:** PT-141 works directly on the central nervous system to boost sexual arousal, rather than just focusing on blood flow, like some other ED treatments.
- 6. Supports Overall Sexual Health:** By improving sexual desire and performance, PT-141 enhances sexual well-being and overall satisfaction with sexual experiences.
- 7. Aids in Managing Low Libido:** PT-141 is effective for individuals struggling with low libido, particularly those whose sexual desire has diminished due to hormonal imbalances, stress, or other factors.
- 8. Improves Sexual Satisfaction:** By enhancing arousal and improving sexual function, PT-141 helps individuals achieve greater sexual satisfaction, improving relationships and quality of life.

SEXUAL FUNCTION



KISSPEPTIN

Lyophilized Powder
SQ Injectable

Kisspeptin

Kisspeptin-10 is a decapeptide that plays a critical role in the regulation of the reproductive system by stimulating the release of gonadotropin-releasing hormone (GnRH) from the hypothalamus. This action triggers the release of luteinizing hormone (LH) and follicle-stimulating hormone (FSH), which are essential for reproductive function. Clinically, Kisspeptin-10 is being investigated for its potential to treat reproductive disorders such as hypogonadism, infertility, and delayed puberty. Its ability to modulate the hypothalamic-pituitary-gonadal axis makes Kisspeptin-10 a promising therapeutic agent for restoring normal reproductive function.

Suggested Dosing Information

100mcg - 200mcg SQ daily

Benefits of Kisspeptin

1. Supports Reproductive Health: Kisspeptin stimulates the release of gonadotropin-releasing hormone (GnRH), which is essential for proper reproductive function and hormone regulation.
2. Enhances Sexual Function: By influencing the secretion of reproductive hormones like luteinizing hormone (LH) and follicle-stimulating hormone (FSH), kisspeptin supports enhanced sexual function in both men and women.
3. Aids in Menstrual Cycle Regulation: Kisspeptin helps regulate the menstrual cycle by ensuring proper hormonal balance, making it particularly useful for women experiencing irregular periods.
4. Improves Fertility: Kisspeptin plays a key role in the regulation of ovulation and sperm production, improving fertility in both men and women.
5. Manages Hormonal Imbalances: It helps to restore balance in the reproductive hormone system, aiding in conditions such as low testosterone in men or irregular hormone levels in women.
6. Reduces Signs of Testicle Atrophy: In men, kisspeptin can prevent or reverse testicular atrophy by supporting natural testosterone production and maintaining reproductive tissue health.
7. Aids in Hypothalamic Disorders Treatment: Kisspeptin is effective in treating hypothalamic disorders like hypothalamic amenorrhea.
8. Enhances Libido: Kisspeptin has been shown to enhance sexual desire and libido in both men and women by optimizing hormone levels.